

- 4 (a) Twenty million years ago there was an ocean between the continents of North America and South America. The snapping shrimp, *Alpheus sp.*, lived in this ocean.

Figure 4.1 shows a snapping shrimp.



Figure 4.1

About 3 million years ago, volcanic activity and sedimentation formed a narrow strip of land, Panama, joining North America and South America.

Figure 4.2 shows the area 20 million years ago and now.

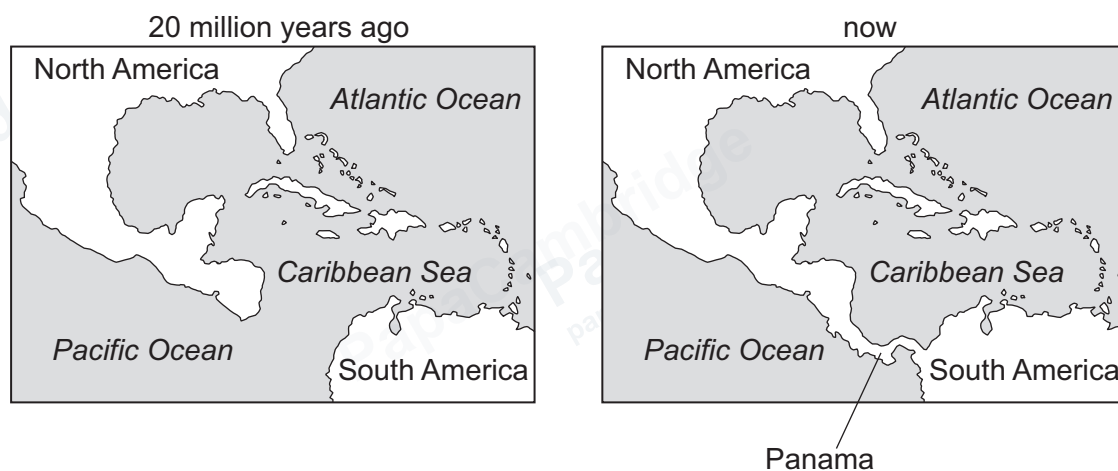


Figure 4.2

Explain how, over time, the snapping shrimp living in the Caribbean Sea and the snapping shrimp living in the Pacific Ocean developed so that they now do **not** mate with each other.

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